

Introductory Matrix Operations

Matrix Entries

Defn. For matrix A , notation a_{ij} means the entry in row i and column j of A .

Matrix Addition and Scalar Multiplication

Defn. Matrix **addition** requires the two matrices have the same dimensions. The sum is defined by adding corresponding entries. Similarly, **scalar multiplication** is defined entry-wise.

For example,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{bmatrix}$$

Matrix Transpose

Defn. The **transpose** of a matrix A , denoted A^T , exchanges rows and columns. That is, $(A^T)_{ij} = A_{ji}$.

For example: here is a matrix and its transpose

$$\begin{bmatrix} 3 & 4 & 7 \\ -2 & 5 & -3 \end{bmatrix} \quad \begin{bmatrix} 3 & -2 \\ 4 & 5 \\ 7 & -3 \end{bmatrix}$$

Square Matrices

Defn. A **square** matrix has equal number of rows and columns.

Diagonal Matrices

Defn. The **diagonal** of a square matrix runs from top-left to bottom-right. A **diagonal matrix** is a square matrix that has zeros off the diagonal (and might or might not have zeroes on the diagonal).

For example

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -\pi \end{bmatrix}$$

Symmetric Matrices

Defn. A ***symmetric matrix*** is a square matrix that is symmetric around its diagonal. In other words, $A = A^T$.

Summary

Matrix addition and scalar multiplication is defined entry-wise. For matrix A , notation a_{ij} means the entry in row i and column j of A .

The transpose of a matrix exchanges rows and columns. A square matrix has equal number of rows and columns. The diagonal of a square matrix runs from top-left to bottom-right. A diagonal matrix has zeros off the diagonal. A symmetric matrix equals its transpose.